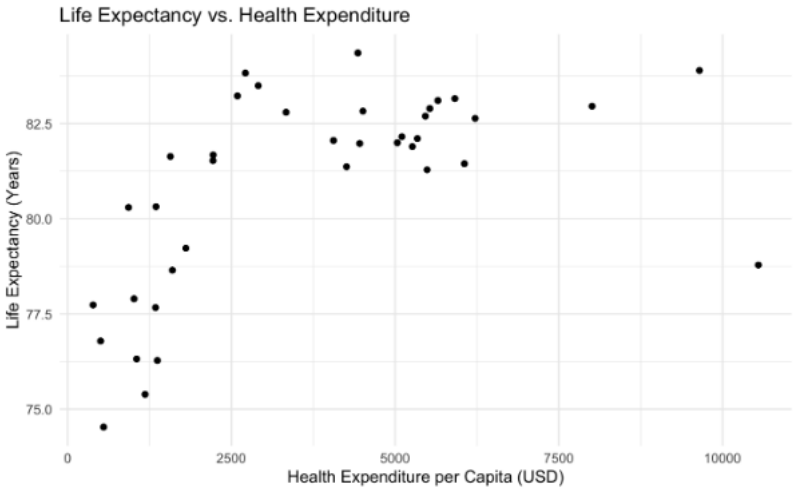
STATS_HW_2

PART1

Question1



Question2

```
Call:
lm(formula = life_exp ~ health_exp, data = health)

Residuals:
    Min      1Q  Median      3Q     Max 
-6.1589 -0.9122  0.3145  1.1012  3.5682 

Coefficients:
              Estimate Std. Error t value             Pr(>|t|)    
(Intercept) 78.6379444  0.6342705 123.982 < 0.0000000000000002 ***
health_exp   0.0005984  0.0001413   4.235             0.000151 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.179 on 36 degrees of freedom
Multiple R-squared:  0.3326,	Adjusted R-squared:  0.314 
F-statistic: 17.94 on 1 and 36 DF,  p-value: 0.0001509
```

- Intercept$\hat{\beta}_0$ ~78.64. This represents the predicted life expectancy when health expenditure is $0.
- Slope$\hat{\beta}_1$ ~0.0006. This means for every $1 increase in health expenditure, life expectancy increases by 0.0006 years.
- Significance: The p-value is ~0.00015（< 0.05）, so the coefficient is statistically significant.

Question3

- At $3,000: Predicted life expectancy is approximately 80.43 years.

- At $6,000: Predicted life expectancy is approximately 82.23 years.

Question4

No. This regression likely suffers from Omitted Variable Bias (OVB). Factors like GDP, diet, or education affect both spending and life expectancy. Without controlling them, the estimate is biased.

Question5

```
Call:
lm(formula = life_exp ~ health_exp + gdp_pc, data = health)

Residuals:
    Min      1Q  Median      3Q     Max 
-4.5219 -1.0255  0.2585  1.3054  3.6549 

Coefficients:
               Estimate  Std. Error t value            Pr(>|t|)    
(Intercept) 78.17775908  0.67294286 116.173 <0.0000000000000002 ***
health_exp   0.00017055  0.00028398   0.601              0.5520    
gdp_pc       0.00005152  0.00002991   1.722              0.0938 .  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.122 on 35 degrees of freedom
Multiple R-squared:  0.3847,	Adjusted R-squared:  0.3496 
F-statistic: 10.94 on 2 and 35 DF,  p-value: 0.0002036
```

- Intercept$\hat{\beta}_0$ ~78.18.
- Health Exp Coefficient $\hat{\beta}_1$ ~0.00017.
- Significance: The p-value for health_exp is ~0.55 （> 0.05）, which is not statistically significant.
- Conclusion: After controlling GDP, there is no evidence that health spending increases life expectancy. The previous positive effect was likely due to GDP.

Question6

```
[1] 0.3140292
[1] 0.349558
```

- Model (2) Adj.R^2: ~0.314.
- Model (5) Adj. R^2~0.350.

- Discussion: The model fits improved after adding GDP, meaning GDP helps explain more variation in life expectancy.

Question7

```
                 2.5 %  97.5 %
health_exp -0.4059623 0.74706
```

- Interval: [-0.41, 0.75] (years).
- Conclusion: Since the interval includes 0 and the upper bound is small (< 1 year), we can reject the hypothesis of a large positive effect.

PART2 Question1

Answer: Incorrect. Residuals being uncorrelated with X is a mechanical property of OLS (always true by construction). It does not prove that the *error term* is uncorrelated with X (no Omitted Variable Bias). Zero correlation of residuals is a **computation result**, not a **causal proof**.

Question2

Incorrect. The slope depends on the variance of the independent variable ($Var(X)$ vs

$Var(Y)$). Unless variances are identical, slopes differ.

Question3

Correct. Higher variance in X (SST_x) lowers the standard error of the OLS estimator, leading to more precise estimates.